

Aaryan Lath

Systems and Design Engineer | (312) 838-5669 | aaryanlath05@gmail.com
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EDUCATION

Purdue University

Bachelor of Science in Aeronautical and Astronautical Engineering (GPA: 3.74/4.0)

Minor in Business Economics, Specialization in Aerodynamics and Systems Design

Honors: Dean's List and Semester Honors for every semester, **AAE Ambassador**

West Lafayette, IN

Expected May 2026

SKILLS

Core Tools: CAD, NX, CREO, Teamcenter, Ansys, Product Design, JIRA, Fluid Mechanics, CAM, SAP

Programming & Analysis: Python, MATLAB, Trade Studies, CFD, FEA, Financial Analysis, HTML, Java

Soft-skills: Analytical, Communicator, Detail-oriented, Fast-paced learner, Inquisitive

Awards: Accounting Case Competition 3rd place, 2nd place in Nationals of World Robot Olympiad Junior High

PROFESSIONAL EXPERIENCE

Zucrow Laboratories (High Speed Compressors Lab)

Undergraduate Research Assistant

West Lafayette, IN

Aug 2025 – Present

- Researching into the inlet vortex distortion phenomena and their effects on turbomachinery performance.
- Helped assemble and calibrate the test cell's fan-rig using laser alignment, reducing time taken by **15%**.

Siemens USA

Mechanical Engineering Intern

Grand Prairie, TX

May 2025 – Aug 2025

- Designed custom enclosures for panelboards using **CREO** and executed ECNs in **SAP**.
- Streamlined **100+** switchboard neutral assemblies configurations by resolving edge case designs.
- Automated BOM generation with Python, cutting manual processing time by **~30%** for **10+** orders/day.

School of Aeronautics and Astronautics

Undergraduate Teaching Assistant

West Lafayette, IN

Jan 2025 – Present

- Led study sessions to teach **50+** students, core **AAE 251** (Aircraft and Spacecraft Design) course material.
- Provided personalized support to students with concepts and **MATLAB** troubleshooting.
- Guided students in understanding key design principles for the aircraft and spacecraft design project.

Resilient Extraterrestrial Habitat Engineering

Undergraduate Research Assistant - Systems Engineer

West Lafayette, IN

May 2024 – May 2025

- Analyzed **18** potential disruptions scenarios in a lunar habitat through conducting **Simulink** simulations.
- Performed trade studies to evaluate safety controls, optimizing **resilience** and costs using **ESM** analysis.
- Executed **FEA** simulations and tests to validate **vibration isolation** devices under various conditions.

PROJECTS AND INVOLVEMENTS

SAE Aero Design

Chief Engineer/ Program Manager

West Lafayette, IN

May 2025 – Present

- Authored a **~75 page** RC aircraft design guide to accelerate onboarding
- Documented past failures from the SAE Aero Design Regular Class to avoid recurring errors.
- Led technical development for **3** sub teams by reviewing trade studies and ensured milestones are met.

Purdue Space Program – A SEDS Chapter

Technical Director

West Lafayette, IN

Dec 2024 – May 2025

- Optimized the clubs backend operations and handled 9 sub-teams for smooth project completion.

Satellites Deputy Systems Director

Feb 2024 – Nov 2024

- Improved project progress with **~15** engineers and other sub-teams ensuring alignment with project goals.
- Formulated, reviewed, and refined **200** project and system-level requirements for Boiler Bus.
- **Manufactured** the satellite frame end plates using **CAM** and walls through **laser cutting**.

Single Piston Sterling Engine

Test Engineer

West Lafayette, IN

Oct 2024 – Jan 2025

- Tested for end-to-end connectivity between **NX** and **Teamcenter** for students in preparation for deployment.
- Reported bugs in Engineering Change Orders (**ECOs**) and Product Data Management (**PDM**) workflow.